Hospital Readmission Prediction

A machine-learning analysis of 10 years of US hospital records for diabetic patients

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github.com/mbhatti48/Data-Project/hospital-readmission-prediction

1. OBJECTIVE

About **1 in 9** hospital discharges of a diabetic patient in this dataset (11.4%) was followed by another hospital admission within 30 days. Readmissions are expensive for hospitals and hard on patients, so this project asks one question: **can we tell, at the moment of discharge, which patients are most at risk of coming back?**

The intended use shapes every choice below: a **screening tool** that ranks discharges for follow-up (a call within the week, an earlier appointment), not a clinical instrument that decides anything about an individual.

2. DATA AND PREPARATION

The analysis uses the 10 Years Diabetes Dataset: ~102K inpatient encounters of diabetic patients across **130 US hospitals, 1999–2008**, with 50+ features per encounter (demographics, admission details, lab results, medications, diagnoses).

Missing values hide behind a literal "?" in this file — 192,849 cells were converted to real missing values before any decisions. Three fields were dropped for missingness or irrelevance (**weight**, ~97% missing; **payer_code**; **medical_specialty**, ~49% missing across 80+ sparse values). The two lab fields load as blank when the test was simply not run, so "not measured" became an explicit category rather than a gap. Encounters that cannot be readmitted — the patient died or was discharged to hospice (2,423 encounters) — were removed, leaving **99,340 encounters from 69,987 patients**.

The raw outcome has three values (not readmitted / after 30 days / within 30 days) and was collapsed to the standard hospital-quality benchmark: **readmitted within 30 days (1) vs everything else (0)**. A readmission at day 45 counts as 0.

3. EXPLORATORY ANALYSIS

Three patterns stood out, and each one shaped the model:

- **Prior hospitalizations are the strongest simple signal.** Encounters with 3+ inpatient stays in the previous year end in readmission 26.4% of the time — three times the rate of patients with none (8.6%).
- **Discharge destination carries a clean gradient.** Patients sent home come back least (9.3%); patients sent to another facility the most (16.3%). Not the discharge causing readmission — sicker patients get sent to facilities — but it is known on discharge day, which makes it fair game for a discharge-day model.
- **Age has two peaks, not a smooth climb.** Risk spikes for young adults in their 20s (14.3%, the highest band — consistent with unstable type-1 disease) and rises again into the 80s.

4. DESIGN: THE LEAKAGE TRAP AND THE PATIENT-GROUPED SPLIT

Thousands of patients appear multiple times in the data — 16,341 patients have 2+ stays, one has 40. A random row split would put the **same patient on both sides**, quietly grading the model on people it trained on and inflating every metric. Discarding repeat encounters kills the leakage but throws away ~30% of the data — and specifically the repeat-visit patients a readmission model most needs to learn from.

The design used here keeps every encounter and closes the trap at the split: the 80/20 train/test split **and every cross-validation fold are grouped by patient**, so all of a patient's stays land on one side only. The notebook verifies the guarantee directly: zero patients appear in both train and test.

Keeping repeat encounters also unlocks **patient-history features**, built strictly from each patient's earlier stays: how many prior stays are on record, whether the previous stay already ended in a 30-day readmission, and the previous and cumulative length of stay. Nothing from the future touches any row; encounter order stands in for time, since the dataset carries no admission dates.

5. MODEL DESIGN

25 features went in: the numeric counts, an ordinal age, the grouped admission/discharge/source codes, the primary diagnosis mapped from 700+ ICD-9 codes to nine body-system chapters, four medication features compressed from 23 per-drug columns, and the four history features above. Categories were one-hot encoded and numeric columns scaled inside a single pipeline, so the scaler is fit on training folds only — nothing from the test set can leak into preparation.

The metric priority was set before modeling: **AUC first** (ranking quality, threshold-free), recall on the readmitted class at a chosen operating point second, accuracy last — with an 89/11 split, "nobody comes back" scores 89% and is useless. Class imbalance was handled with balanced weights, so every row stays in training. Four candidates were compared by patient-grouped cross-validated AUC, and the leader was tuned by cross-validated grid search over learning rate and tree depth.

6. MODEL EVALUATION

Model	Cross-validated AUC	Why it matters here
Always predict the base rate	0.500	The no-skill floor
Logistic regression	0.658	Interpretable baseline; the bar to beat
Random forest	0.642	Bagged trees — trails the linear model here
Gradient boosting	**0.665**	**The leader; tuned and carried forward**

Patient-grouped 5-fold cross-validation on the training set. The linear model running this close to the tree ensembles says the signal is mostly additive — which caps what tuning can add.

Scored once on the held-out test set, the tuned gradient-boosting model reached **test AUC 0.672** (PR-AUC 0.215 against a base rate of 0.113); the logistic baseline 0.665. Readmission is genuinely hard to predict — published results on this dataset land in the same band, and results far above it should raise suspicion of leakage, not applause.

AUC measures ranking; using the model requires a threshold. For screening, missing a readmission costs more than a false alarm, so the threshold was lowered deliberately:

Decision threshold	Recall	Precision	Share of discharges flagged
0.550	0.50	0.19	~29%
0.509 (chosen)	**0.60**	**0.18**	**~38%**
0.459	0.70	0.17	~48%

Test-set operating points for the tuned model.

At the chosen point the model **catches 3 in 5 readmissions** while flagging ~38% of discharges, most of them false alarms. That is a wide-net screening trade: appropriate when follow-up is cheap (a phone call), inappropriate when it is not. The model ranks; the economics pick the threshold.

7. WHAT DRIVES READMISSION RISK

The logistic model's odds ratios and the boosted model's feature importances were read side by side. Two very differently built models ranked the same signals at the top — strong evidence the signal is real.

Feature	Odds ratio (logistic)	GB importance	Effect on risk
Inpatient visits, prior year	1.33	0.43	Raises — the dominant signal in both
Discharged: home	0.73	0.09	Protects, the strongest safeguard
Discharged: another facility	1.31	0.11	Raises
Previous stay ended in readmission	1.24	–	Raises (history feature)
Cumulative prior days in hospital	–	0.09	Raises (history feature)
On any diabetes medication	1.28	low	Raises — likely a severity marker
Age	–	0.03	Modest; two-peak shape
Respiratory / musculoskeletal diagnosis	0.85–0.87	low	Protects (vs other primary diagnoses)

Odds ratios above 1 raise the odds, below 1 lower them; numeric features are per standard deviation. Importances are the share of the boosted model's split decisions and carry no direction.

The profile: the high-risk discharge belongs to a patient who keeps being hospitalized — recent inpatient visits, prior stays on record, a previous stay that already ended in readmission — who is not going home, and who is older or carries a circulatory primary diagnosis. Nothing exotic outranks that simple story: no lab value or specific medication came close.

8. WHAT HELPED, AND WHAT DID NOT

The project's biggest gains came from **data-design decisions, not algorithms**. A first-encounter-only version of this analysis (repeat visits discarded) topped out at test AUC 0.645. Keeping all encounters behind the patient-grouped split added more than +0.02; the history features a further +0.004 (cross-validation and test agreed on both). By contrast, a much wider hyperparameter search, extra features (grouped secondary diagnoses, admitting specialty, dosage-change counts), and blending the two models' rankings bought about **0.002 AUC combined**. On this dataset, the ceiling belongs to the design and the features — not to the algorithm or its settings.

9. LIMITATIONS

- The data is old (1999–2008) and American; diabetes care has changed, and nothing transfers to another health system without revalidation.
- AUC 0.67 is modest — it supports prioritizing follow-up, not clinical decisions about individuals.
- The dataset has no admission dates; encounter order stands in for time in the history features — standard practice for this dataset, but an assumption.
- Repeat encounters are not independent: frequent flyers contribute more rows, so the model leans toward the repeat-visit population — which is also the population a screening tool most needs to serve.
- Precision at the chosen threshold is ~18%: most flags are false alarms, tolerable only when follow-up is cheap.
- One dataset, one split, no external validation — the honest next step is a different hospital system's data.

10. CONCLUSION

The question is answered, with the scope stated. **Readmission risk can be ranked, not pinpointed:** a tuned gradient-boosting model reaches test AUC 0.672 in a fully patient-leakage-free evaluation, and used as a wide-net screen it catches 3 of 5 readmissions. **What drives the risk is hospitalization history in every form** — recent visits, prior stays, a previous readmission — followed by discharge destination, with two unrelated models agreeing on the ranking. And the project's clearest methodological lesson: when the score stalls, look at the data design before the hyperparameters — the grouped-split design was worth ten times what all the tuning combined delivered.

11. TOOLS AND SOURCES

Built in Python: pandas and NumPy for the data work, Matplotlib for visualization, and scikit-learn for preprocessing, modelling, and evaluation. The annotated notebook ships in the repository; the dataset is not included due to size (the README links the download).

Data: 10 Years Diabetes Dataset (Kaggle), derived from the "Diabetes 130-US hospitals for years 1999–2008" clinical database. **Reference:** Strack, B. et al. (2014), "Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records," *BioMed Research International* — source of the dataset documentation and the comparable published performance range.